

Novel Methods of Spacecraft Navigation using X-ray Pulsars

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SCPNT 2022
October 13, 2022



Outline

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- Background to Navigation with Variable Celestial Sources
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- Pulsar Phase Tracking Background
- Pulsar Signal Processing
- Novel 3rd Order Phase Tracking
- Analysis and Simulation Results
- Hybrid Energy Navigation and Horizon Crossing Methods
- Conclusions and Future Work

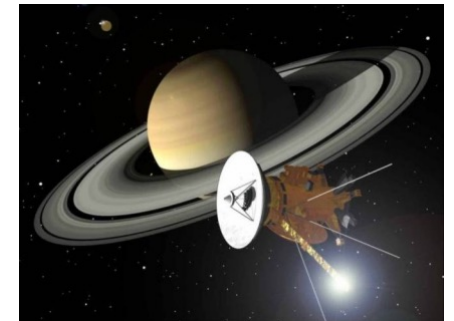
Introduction: Spacecraft Navigation



- GPS (LEO/HEO, GEO)
- Deep Space Navigation
 - DSN
 - 3 radio communication complexes
 - Optical
 - Star tracker
 - Imaging of target planet
- X-ray Pulsar-based Navigation (XNAV)
 - Standard accuracy throughout solar system
 - Potentially autonomous
 - Reduce load on DSN



NASA/CXC/ASU/J. Hester et al.



Images courtesy of NASA

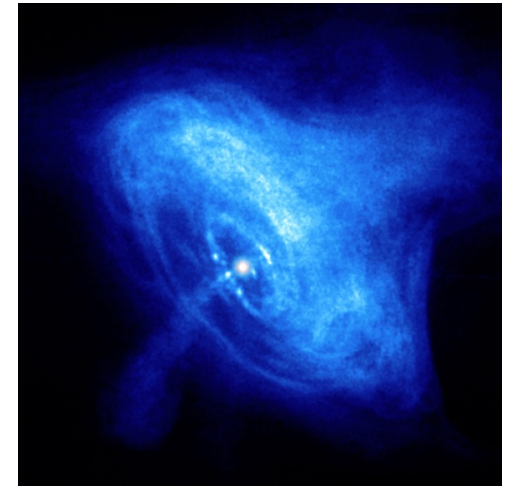
Variable Celestial Source Attributes for Navigation

Source Types	Unique Attributes	Detection Methods and Techniques	Navigation Applications
Radio Pulsar	• Periodic • Numerous • Good geometric distribution	• Large aperture radio antenna or dishes	• Timing • PT scale • Position, velocity
X-ray Pulsar	• Periodic • Generally low flux	• Gas proportional counters • Microchannel Plates • Scintillating materials • CCD and solid-state semiconductors	• Timing • PT scale • Position, velocity • Relative navigation
Gamma-ray Pulsar	• Periodic • Very low flux	• Large volume scintillating materials	• Position, velocity
Bright X-ray Sources	• Aperiodic • Numerous • Many emit high flux	• Scintillating materials • CCD and solid-state semiconductors • Coded aperture masks • X-ray telescopes	• Relative navigation • Attitude
Gamma-ray Bursts	• Singular event • Omnidirectional • Extremely high flux	• Small volume scintillating materials • Semiconductors	• Relative navigation

Introduction to Pulsars

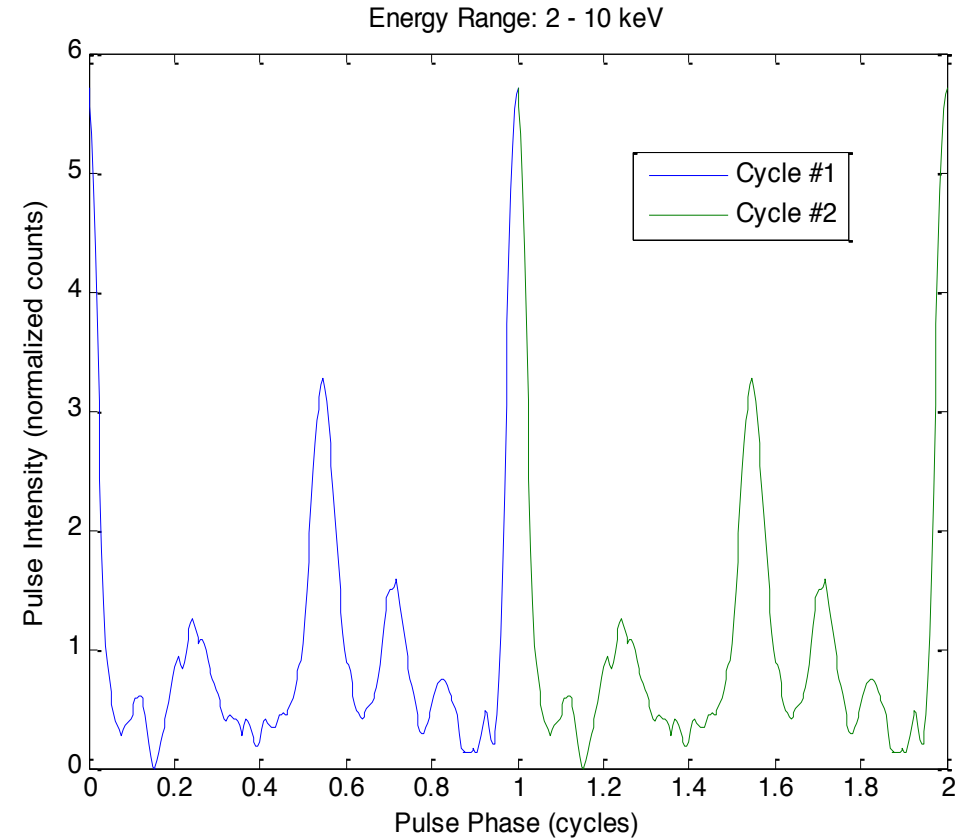
- Spinning neutron stars
- Discovered in radio band in 1967, Hewish and Bell
- Emit radiation throughout the electromagnetic spectrum
- Formed as result of supernova
 - Spin up to high rates
- Strong magnetic field which accelerates charged particles
 - Misalignment of spin and magnetic field axes

NASA/CXC/ASU/J. Hester et al.



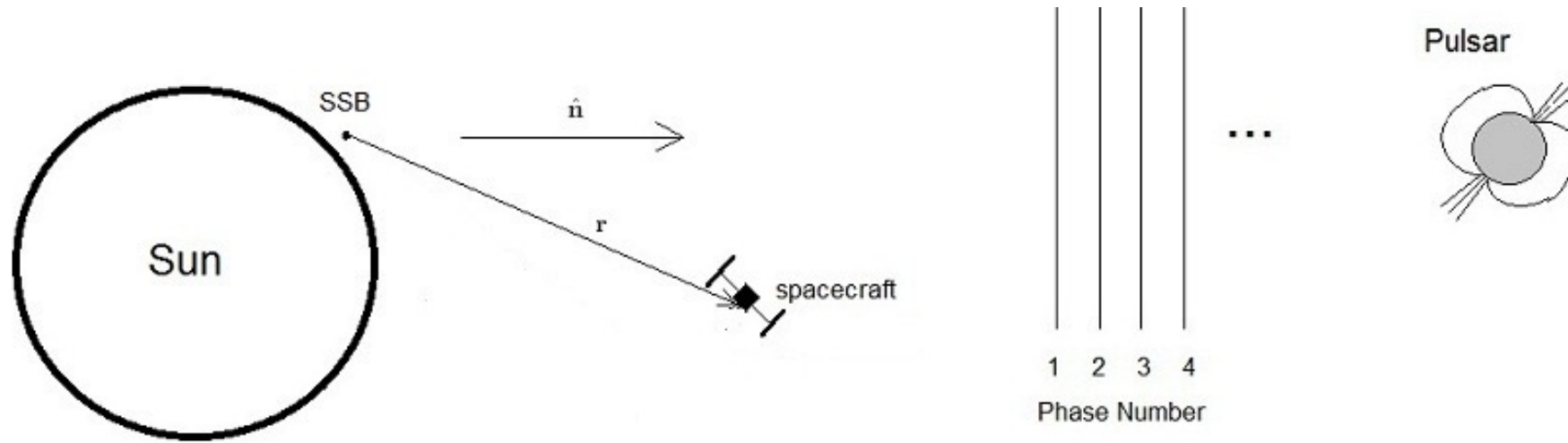
Advantages for Navigation

- Millisecond period pulsars (MSPs)
 - Very stable, reliable signals
- Unique periodic pulse profiles
- Good distribution throughout sky
- Relative position considered fixed
- X-ray band requires smaller detectors
- Large catalog of pulsars
 - Astronomical community



Simulated profile of PSR B1821-24
over two periods

Pulsar-based Navigation



- Celestial “lighthouse”
 - Use each pulsars unique profile to provide position and velocity measurements
- Measurements from 3 pulsars for 3D state

$$\Delta t = \frac{\hat{\mathbf{n}} \cdot \mathbf{r}}{c} + \text{H.O.T.}$$

Selected Historical Contributions

Name	Year	Description
Downs	1974	First to propose radio pulsars for spacecraft navigation
Chester and Butman	1981	Proposed X-ray pulsars for interplanetary navigation
Wallace	1988	Radio pulsars for Earth navigation
Wood	1993	X-ray navigation in near Earth orbit
Hanson	1996	Doctoral thesis on X-ray pulsars for attitude and time
Sala	2004	ESA feasibility study
Sheikh	2005	Doctoral thesis giving extensive XNAV study
Golshan and Sheikh	2007	Previous attempt at phase tracking X-ray pulsar signals
Emadzadeh	2009	Doctoral thesis on relative navigation using X-ray pulsars
Hisamoto and Sheikh	2015	Gamma-ray bursts as aids for spacecraft navigation
Anderson	2021	Doctoral thesis on phase tracking X-ray pulsar signals for navigation
Various from China	Recent Years	Range of XNAV investigations including filter design and modeling

Flight Experiments

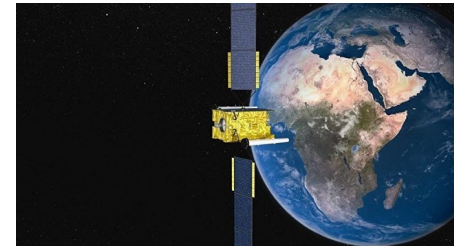
- USA Experient, ARGOS, NRL, 1999
 - Earth orbit attitude determination
- HASP, University of Minnesota, 2012, 2014, 2016, and 2017
 - High altitude balloon experiments
- XPNAV-1, China, 2016
 - Characterized X-ray pulsar signals
- SEXTANT, NASA, 2017
 - Initial real-time demonstration of XNAV on ISS



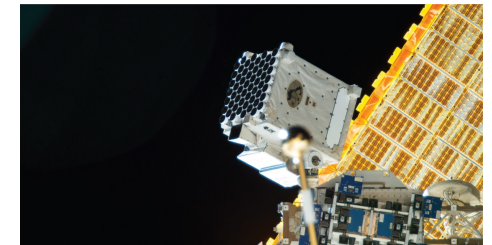
USA Instrument, NRL



HASP, UMN



XPNAV-1, CAST



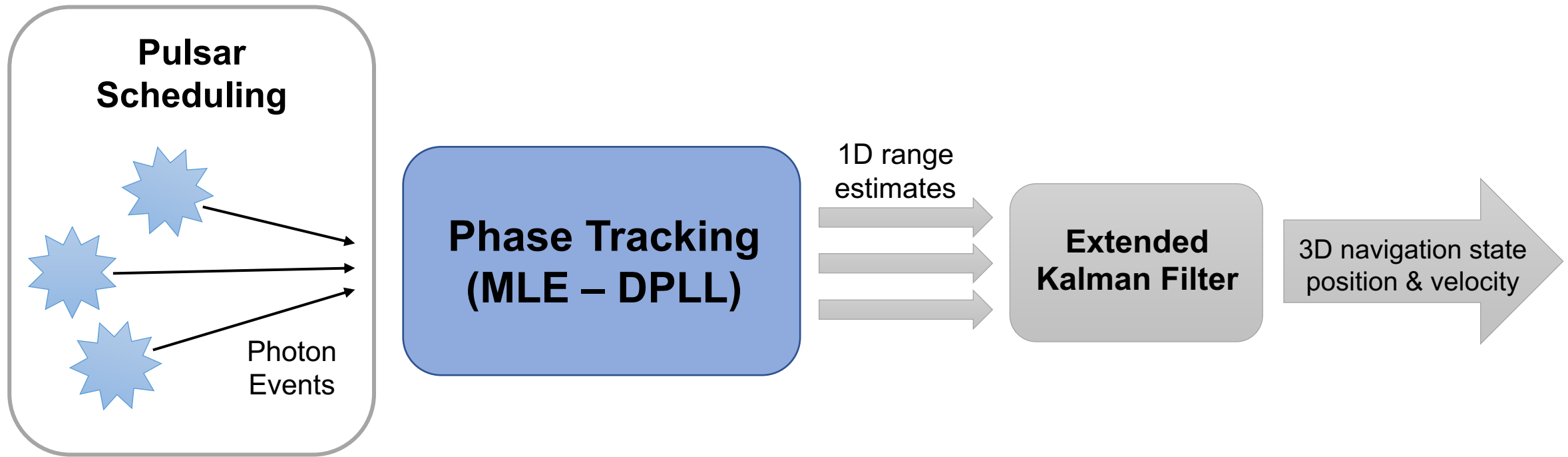
NICER, NASA GSFC

XNAV is a work in progress with challenges being addressed in research

Overview of Deep Spacecraft Navigation Methods

		Pros	Cons
Current State of the Art	DSN	• Good range and range rate estimates (1 km/ 1 AU) • Uses technology available on spacecraft for communication	• Accuracy decreases far from Earth • Requires Earth-based processing • Increased load from future missions
	Binning	• Good single pulse TOA measurement (km level accuracy) • Similar to astronomy approach	• Long observation per estimate (10^4 s) • Information lost in binning • No Doppler frequency estimate
XNAV	2 nd Order Phase Tracking	• Frequent estimates • Estimates phase and Doppler frequency (position and velocity)	• Constant frequency assumption • Requires large detectors for MSPs (>25 m²)
	3 rd Order Phase Tracking	• Allows for changing observed frequency • Estimates phase, Doppler frequency, and frequency derivative • Estimates every 100 to 5000 s	• Requires initial frequency and frequency derivative (acceleration) estimates (handoff from DSN)

Conceptual Flow for Phase Tracking



Analysis of Pulsar Signals

- Source signals are faint
 - Many periods between photon arrivals (cannot just detect signal)
 - Diffuse X-ray background and shot noise from pulsar
- For X-ray detectors that record individual events, Poisson statistics dominate the random noise
- Model photon events as a NHPP $\lambda(t) = R_b + (1 - \rho)R_s + \rho R_s h(\phi_{det}(t))$ (ph/s)
- Observed signal frequency changes with velocity

$$\phi_{det}(t) = \phi_0 + f_s(t - t_0) + \phi_d(t)$$

Previous Phase Tracking (Golshan & Sheikh, 2007)

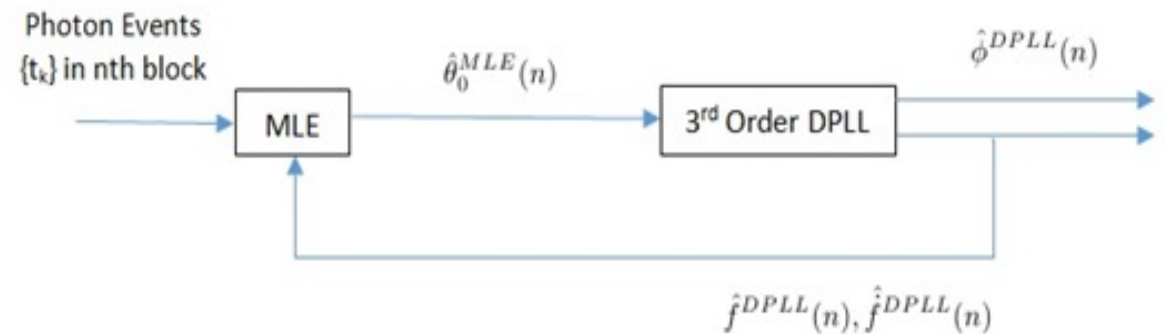
- Partition the observed photon events
 - One second blocks
- MLE for initial phase assuming constant frequency
- Second-Order DPLL
 - Provides Doppler frequency estimate
- Good phase and frequency tracking with Crab Pulsar
- Fails for lower flux pulsars
 - Not enough photons accumulate per block
 - Most of the best pulsars for navigation are orders of magnitude lower in flux than the Crab Pulsar

Proposed Phase Tracking Method

- 2nd order Taylor series for Detector Phase
 - Relax constant observed frequency assumption
 - Longer Block Lengths

$$\phi_{det}(t) = \phi_0 + f(t - t_0) + \frac{\dot{f}}{2}(t - t_0)^2$$

- Estimate of frequency and frequency derivative used to form parabolic model for phase
 - Feedback from a 3rd order DPLL
 - Only unknown is ϕ_0 for MLE
 - Handoff from DSN or GPS

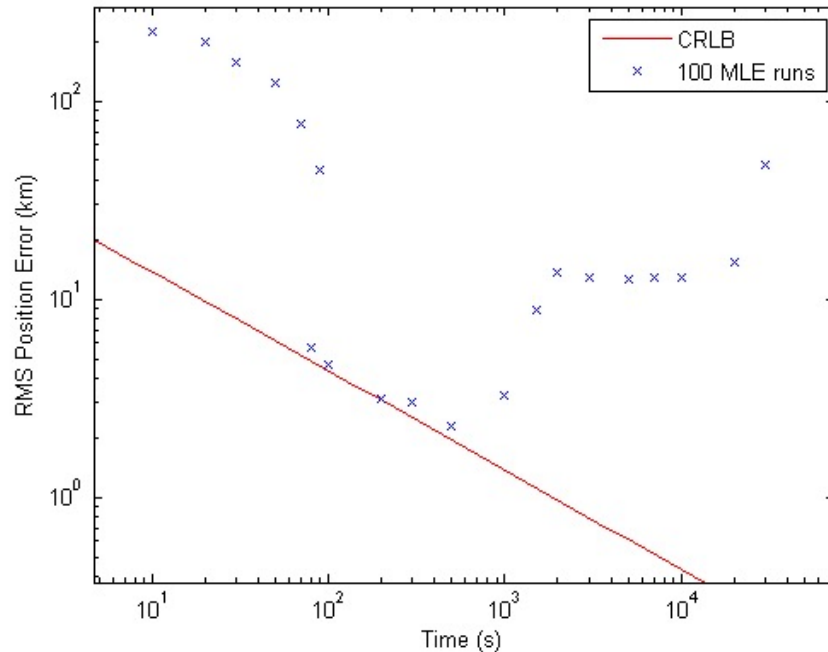


Phase Tracking Simulations (1D range)

- B1821-24, B1937+21, J0218+4232, J0437-4715
 - 4 and 5 orders of magnitude lower flux than Crab
 - Photon times simulated using XPRESS package from ASTER Labs
- Variety of Dynamic Stress Conditions
 - MSL interplanetary cruise trajectory
 - Cislunar transfer trajectory
 - Earth orbits: DirecTV, GPS, and ISS
- Initial condition errors:
 - 1 km, 1 m/s, 1 mg or 1 cm/s² for cislunar and MSL
 - 10 m, 10 cm/s, 1 mm/s² for Earth orbits
- Detector Area: 1 m²

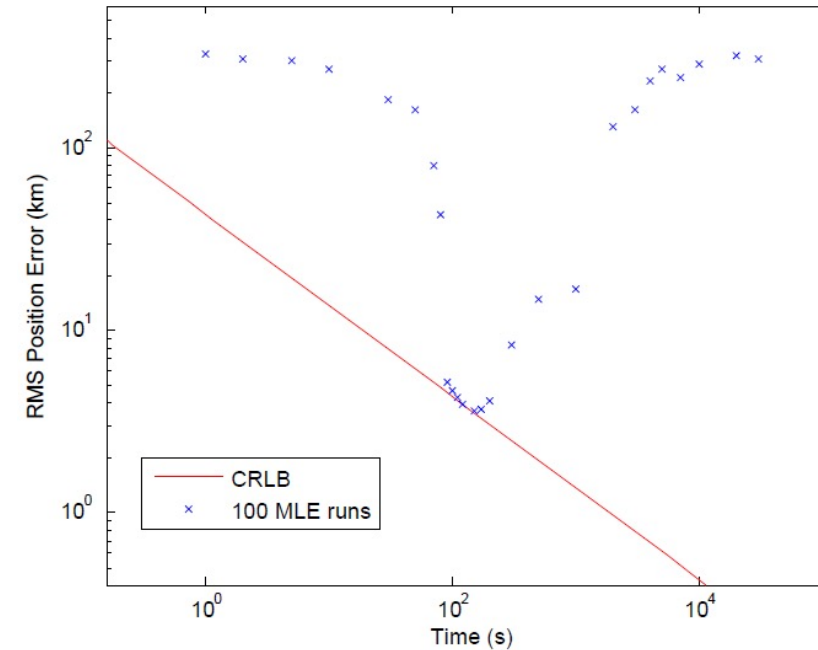
Empirical MLE Simulations with B1821-24 and Orbit Dynamics

GPS Orbit



$$80 \text{ s} < T < 500 \text{ s}$$

ISS Orbit



$$90 \text{ s} < T < 170 \text{ s}$$

Loop Noise Bandwidth Choice

- B_L is the parameter along with T that can tweak DPLL performance
- Trade-off between steady-state tracking error and sensitivity to dynamic stress
 - Smaller B_L leads to smaller steady-state tracking position error
 - Higher B_L allows the DPLL to stay in-lock for higher dynamic stress

Pulsar Parameters

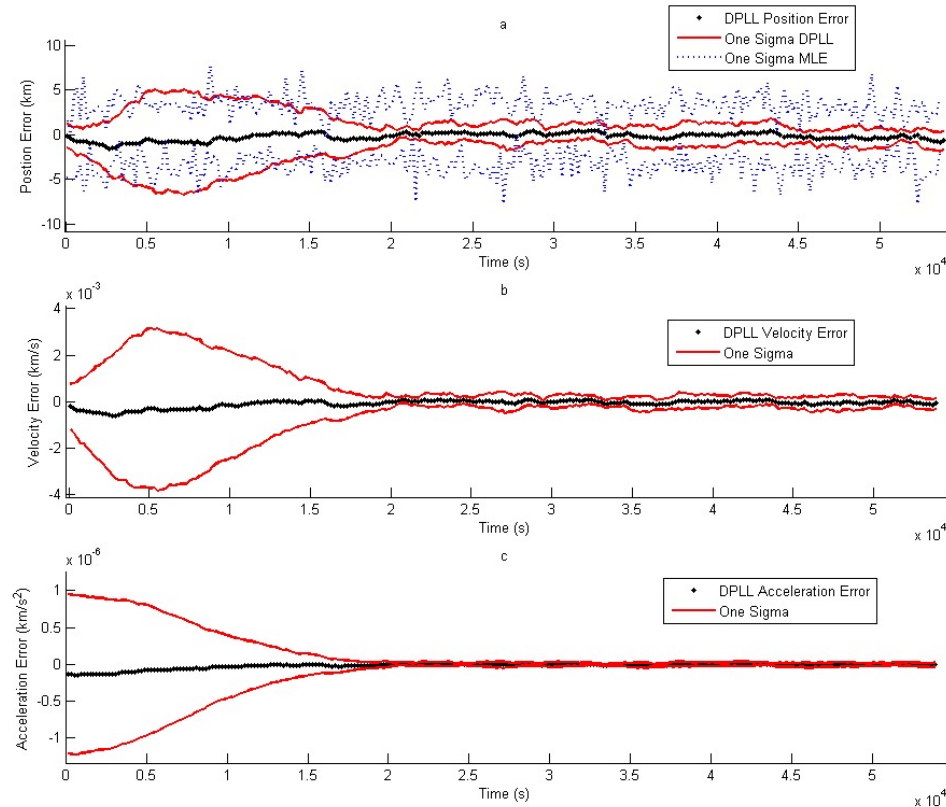
	B0531+21, Crab	B1821-24	B1937+21	J0218+4232	J0437-4715
Period (s)	0.0334	0.00305	0.0015578	0.00232	0.00575
F _s (ph/cm ² /s)	1.54	1.93e-4	4.99e-5	6.65e-5	6.65e-5
SNR (α/β)	2.3283	0.18841	0.04262	0.04769	0.01777
P _f	0.7	0.98	0.86	0.73	0.28
T (s)	10	200	4000	4000	4000

Background Flux $F_b = 0.001$ ph/cm²/s

Detector Area: 1 m²

Phase Tracking Results: B1821-24

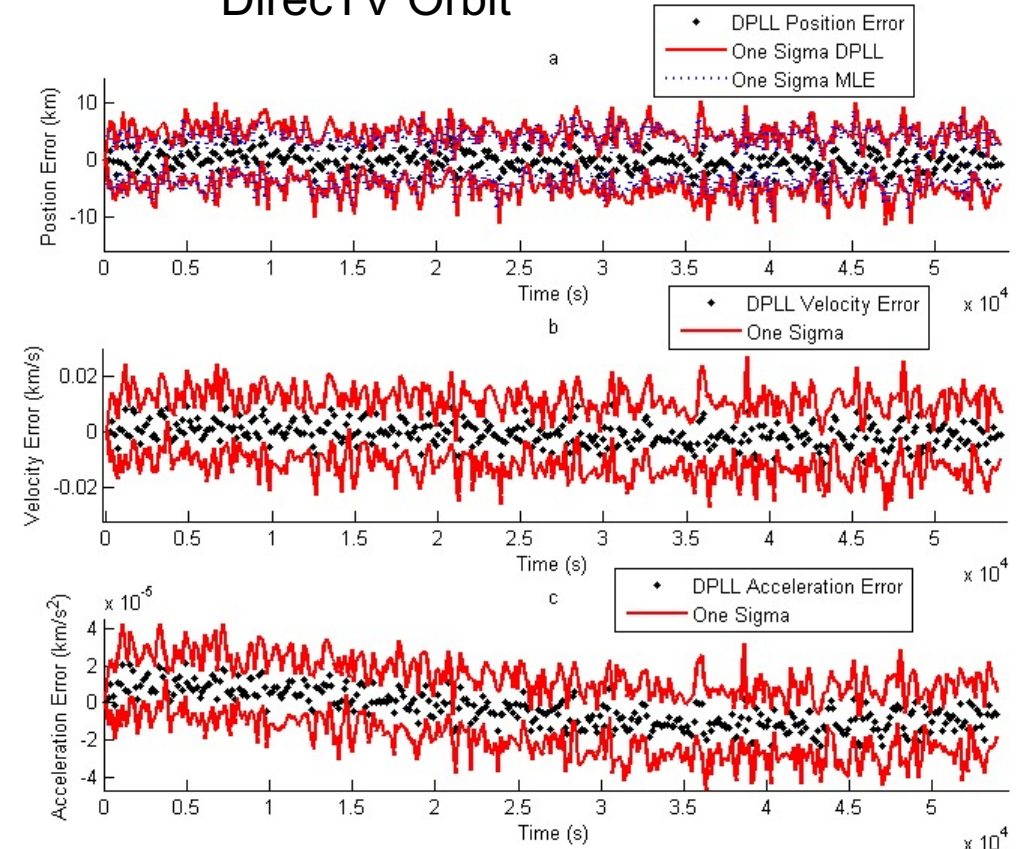
MSL Cruise Trajectory



- Mean Final Position Error: 1.16 km
- Mean Final Velocity Error: 24 cm/s

$B_L = 0.00025$ Hz
 $T = 200$ seconds

DirecTV Orbit



- Mean Final Position Error: 2.47 km
- Mean Final Velocity Error: 6.25 m/s

$B_L = 0.005$ Hz
 $T = 150$ seconds

MSL Cruise Trajectory Results

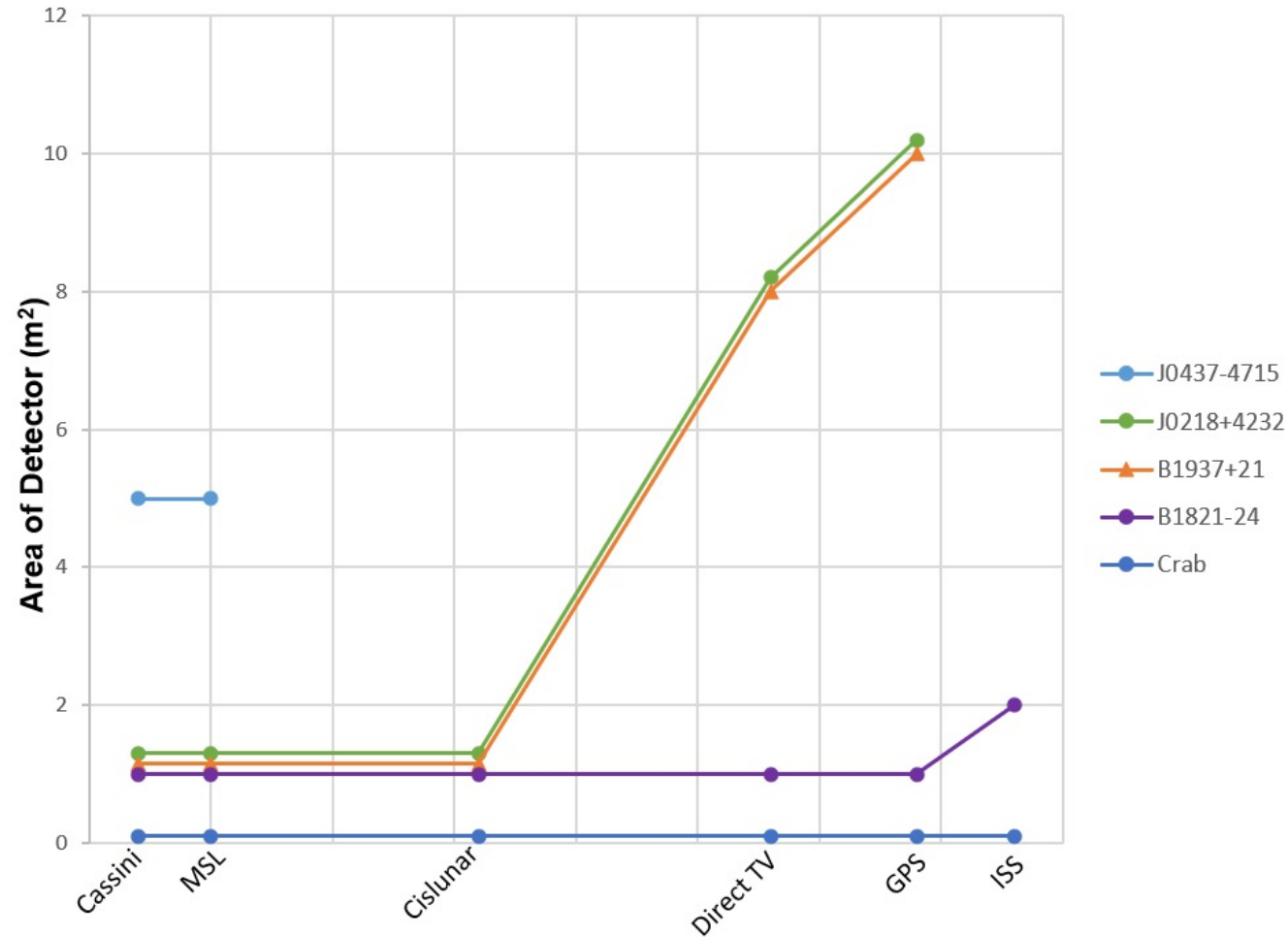
Pulsar	End DPLL RMS Range Error (km)	End DPLL RMS Range-Rate Error (cm/s)	MLE Range Error (km)	CRLB (km)
PSR B1821-24	1.16	24	3.38	3.07
PSR B1937+21	4.12	41	2.53	1.02
PSR J0218+4232	6.45	63	5.42	4.19
PSR J0437-4715 (5 m ²)	36.19	226	20.30	18.49
Crab Pulsar	0.61	53	1.46	1.22

DirecTV Orbit Results

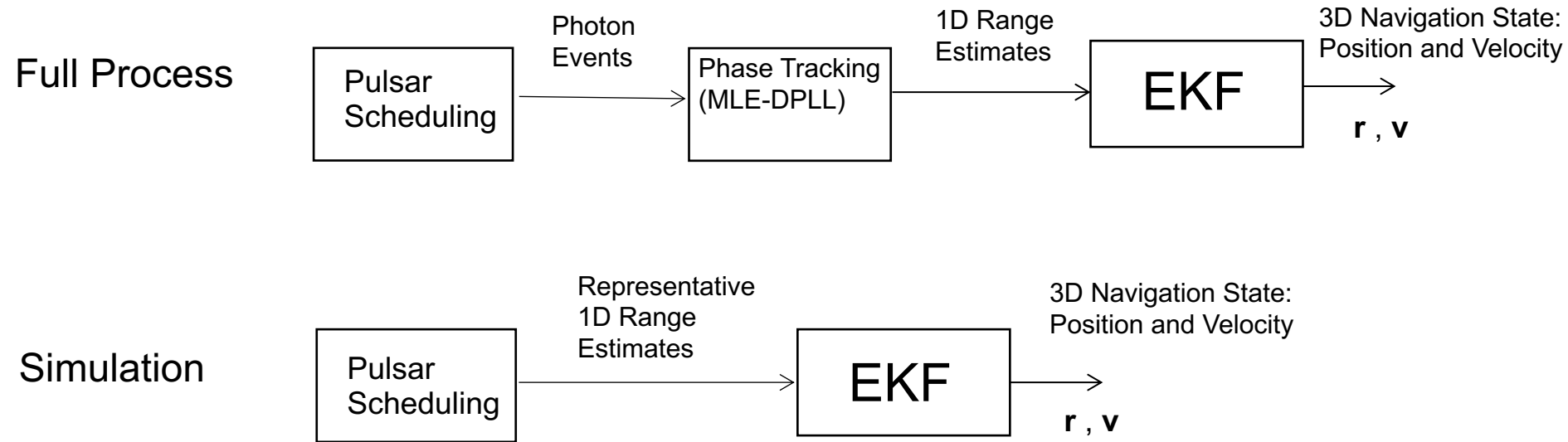
Pulsar	End DPLL RMS Range Error (km)	End DPLL RMS Range-Rate Error (cm/s)	MLE Range Error (km)	CRLB (km)
PSR B1821-24	2.47	6.25	3.93	3.55
Crab Pulsar	0.47	2.08	1.46	1.22

- B1937+21 and J0218+4232 required 8 m²

Detector Areas Needed for Each Pulsar to Phase Track Various Orbits

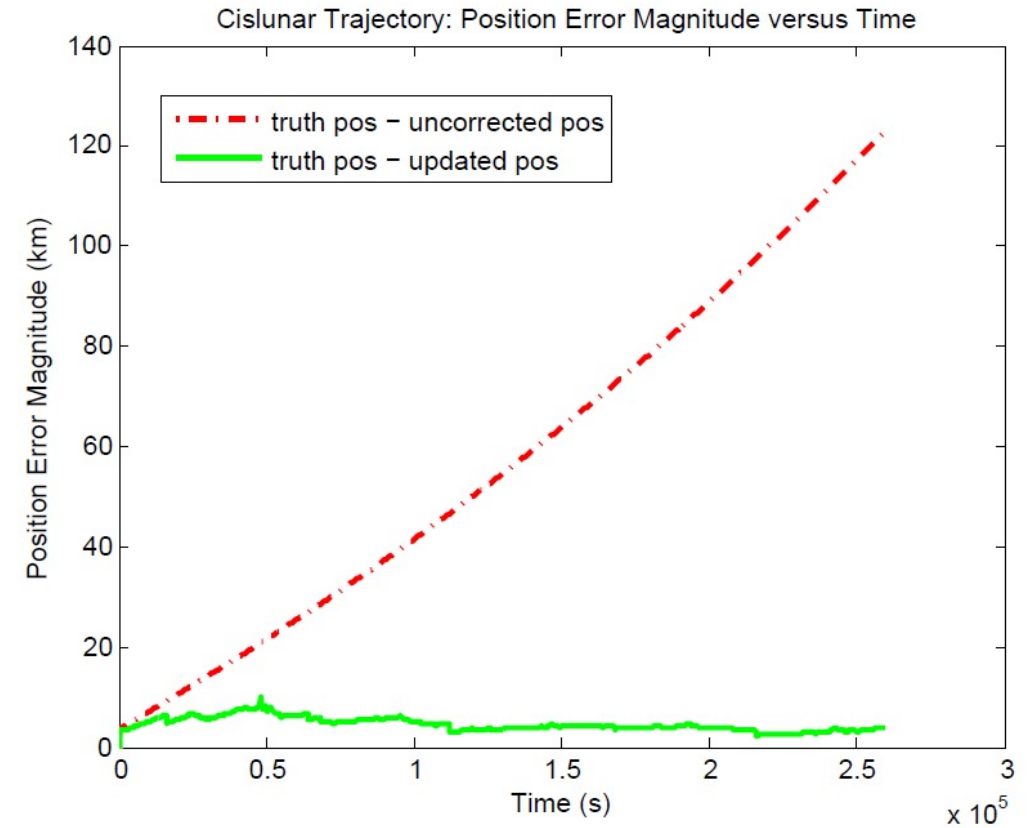
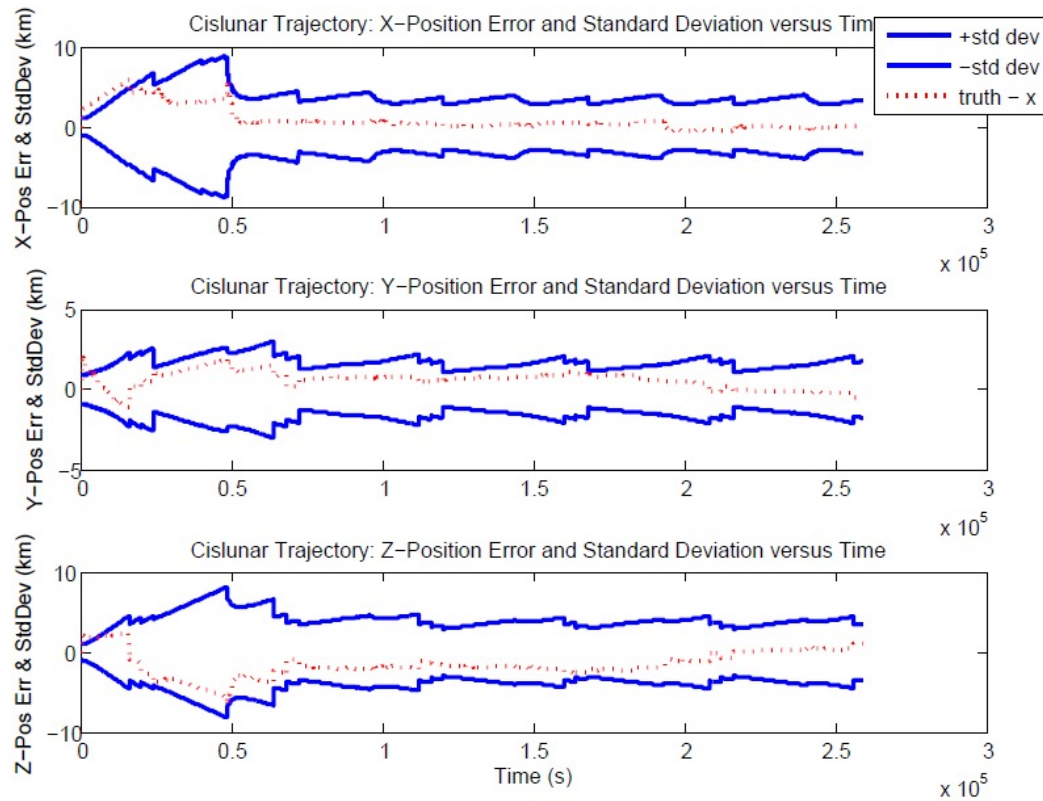


EKF Simulations for 3D State



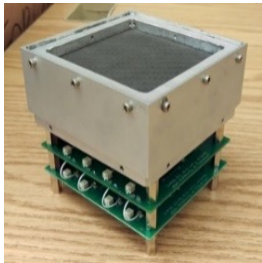
- MSL and Cislunar with 1 m² Detector Area
- Initial Condition Error
 - [2; 2; 2] km and [20; 20; 20] cm/s
- 3 Day Simulation Duration (repeated 10 times)
- Schedule: B1821-24, B1937+21, Crab pulsar, J0218+4232
 - 12000 seconds per pulsar

Cislunar Trajectory EKF Results

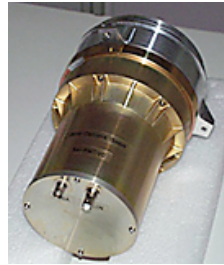


Mean Errors: 3.93 km and 5.49 cm/s

HYNAV Concept

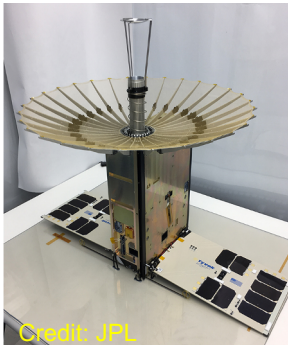


X-ray
Detector



Gamma-ray
Detector

Radio
Dish



16U Microsat

Combine Hybrid Energy
Detectors

Single instrument per Spacecraft

Radio: > 400 MHz

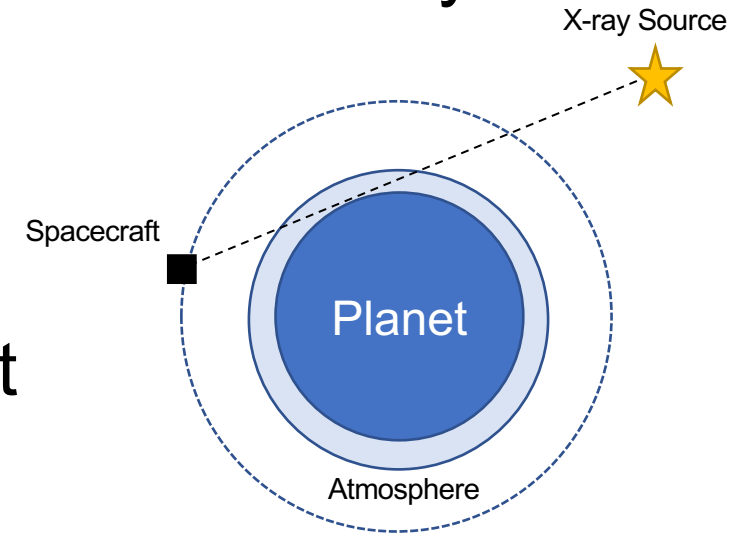
X-ray: 0.1 – 10 KeV

Gamma-ray: > 20 KeV

- Hybrid-energy Navigation
- Combine radio, x-ray, and gamma-ray observations in single instrument
- Advantages from each type while minimizing negatives
- Single or distributed spacecraft
- Autonomous navigation instrument for multiple spacecraft operation
- 100 – 5000 m MRSE with greater robustness to orbit
- Larger source availability
- Improved operating space where top level position errors are achievable

Navigation using Horizon Crossings

- Position information from timing of X-ray source occultation by planet and atmosphere
- Early work by Kent Wood at NRL
- Similar methods in optical
- Accuracy linked to distance from targeted planet
- Can be non-variable point sources
 - Wider range with high flux
- Recent work at Haverford College
 - NICER horizon crossings
- Simulations to combine XNAV and horizon crossing navigation



Future Work

- Internal MLE phase models for Earth orbits
- Photon level EKF simulations
- CubeSat to test phase tracking methods
- Phase tracking advancements to approach single photon processing
 - Nonlinear or suboptimal filtering

Conclusions

- Fertile area of research for navigation with variable celestial sources
- Many benefits including the potential for more autonomy
- Challenges including that many sources are faint
- Phase tracking and horizon crossing measurement are promising areas for further study
- New phase tracking method shown to track phase and frequency for low flux MSPs with detector area $\approx 1 \text{ m}^2$
- Combined 3D EKF
 - 3 km error for MSL and 4 km for cislunar

Acknowledgments

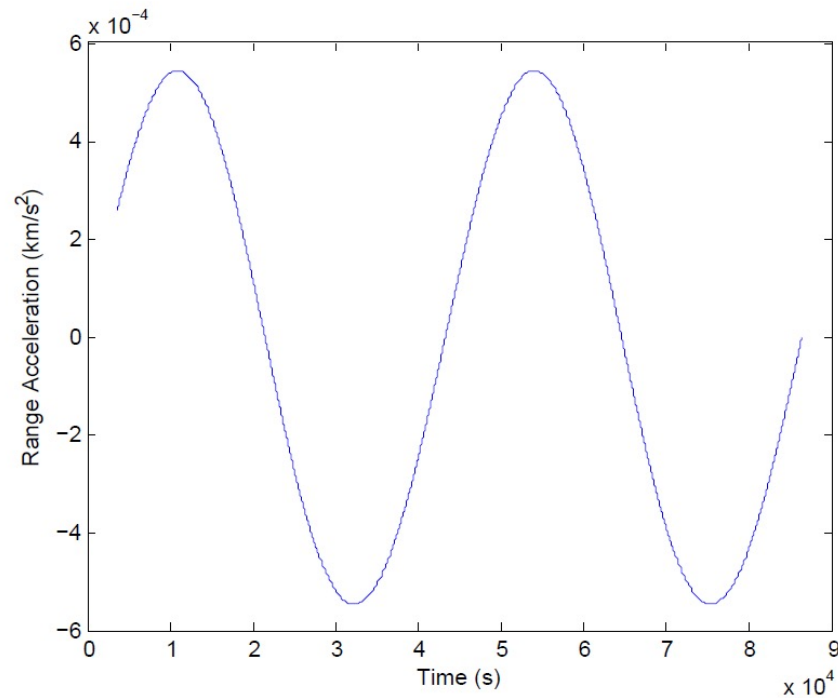
- Dr. Robert Golshan
 - Formerly, The Aerospace Corporation
- Dr. Keith Gendreau and NICER/SEXTANT
 - NASA Goddard
- Nathaniel Ruhl and Dr. Andrea Lommen
 - Haverford College
- Dr. Kent Wood
 - Formerly, NRL
- Chuck Hisamoto and others at ASTER Labs

Questions?

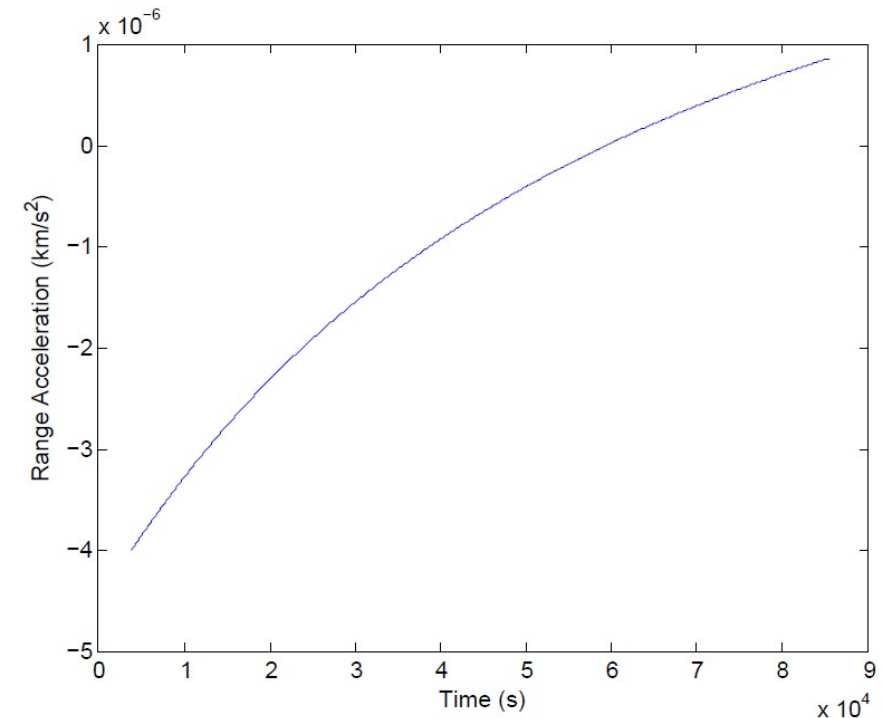
Backup Slides

Dynamic Stress

GPS Orbit



Cislunar Trajectory



- Demonstrates offset from constant acceleration